

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	16
Experiment Name	Understand Pandas Library Use Cases

DESCRIPTION

To perform the experiment: Understand Pandas Library Use Cases.

CODE

```
import pandas as pd

df = pd.DataFrame({
    "Product": ["A", "B", "A", "C", "B"],
    "Sales": [200, 150, 220, 300, 180],
    "Quantity": [4, 3, 5, 6, 4]
})
print("DataFrame:")
print(df)
print("\nFirst rows:")
print(df.head())
print("\nInfo:")
df.info()
print("\nDescription:")
print(df.describe())
print("\nFiltered rows:")
print(df[df["Sales"] > 180])
print("\nGrouped totals:")
print(df.groupby("Product")[["Sales", "Quantity"]].sum())
```

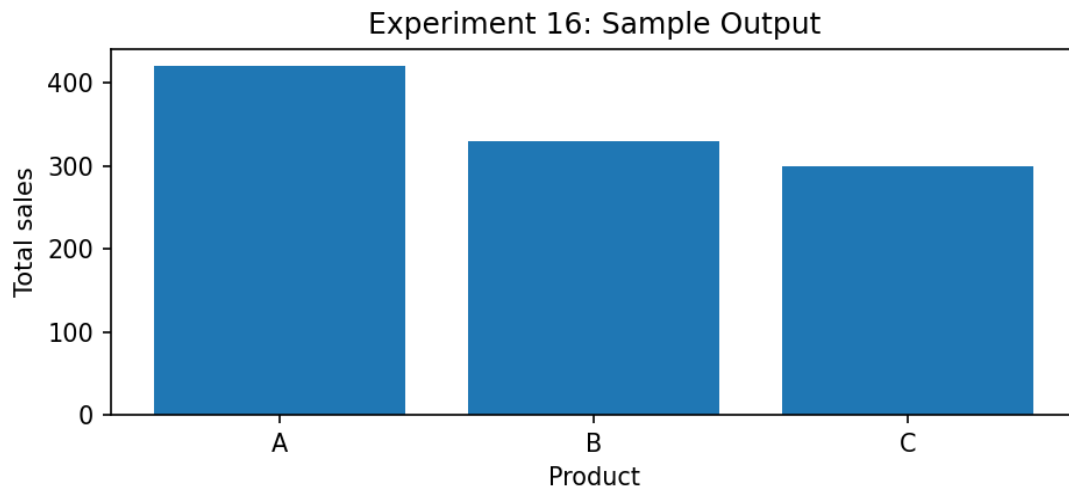
SAMPLE INPUT

Sample input: five product records.

OUTPUT — SAMPLE CONSOLE OUTPUT

Grouped totals:\nA: Sales 420, Quantity 9\nB: Sales 330, Quantity 7\nC: Sales 300, Quantity 6

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature